

A DYADIC OBSTRUCTION, AND K_4 , AGAINST THE THIRDS CONJECTURE FOR CUBIC GRAPHS UNDER UNIFORM RANDOM EMBEDDING

ABSTRACT. Ghanbari and Šámal conjectured that in a random embedding (π, λ) of a bridgeless cubic graph G with m edges, the expected numbers of bad singular, good singular and regular edges are each exactly $m/3$. The source does not define the underlying probability measure; we settle the conjecture under two uniform models of “random embedding” — all signed rotation systems, and the orientable ones — and we do not settle it under any non-uniform or limiting measure. (Sampling signatures at one fixed rotation is not a third model: by the switching lemma below it is another parametrization of the first, with the same induced distribution of edge classes.) Under the uniform measure on the 2^{n+m} signed rotation systems all three parts fail, in two independent ways. First, a divisibility argument: for a connected edge-transitive cubic graph $\text{Aut}(G)$ -equivariance forces each per-edge class probability to be a single dyadic rational $A_i/2^{n+m}$, so $\mathbb{E}[\#\text{class } i] = m/3$ would require $3A_i = 2^{n+m}$. Second, an exact census of the $2^{10} = 1024$ signed rotation systems of K_4 , carried out at one fixed rotation and tabulated signature by signature in the accompanying program, gives $(\mathbb{E}[\#\text{bad}], \mathbb{E}[\#\text{good}], \mathbb{E}[\#\text{reg}]) = (\frac{27}{16}, \frac{69}{32}, \frac{69}{32})$ against the conjectured $m/3 = 2$, so $m/3$ is not a member of that multiset and no relabelling of the three classes rescues any part. On the uniform orientable model the numbers change but the three equalities still fail. What is contradicted is the exact constant $m/3$; the near-uniform split the source observed is not in dispute, and the conjecture’s only consumer in the source paper keeps its conclusion, which that paper already proves unconditionally by a different route.

1. THE CONJECTURE, VERBATIM

Let G be a graph. Following Mohar and Thomassen [9], a *combinatorial embedding* of G is a pair (π, λ) , where π assigns to each vertex a cyclic order of the darts at that vertex and $\lambda : E(G) \rightarrow \{+1, -1\}$ is a *signature*. Ghanbari and Šámal [7, lines 117–120 of `main.tex`] classify an edge e of an embedded graph as follows: e is *singular* if some facial walk F traverses e twice, and *regular* otherwise; a singular edge traversed twice in the same direction by F is *good singular*, and one traversed twice in opposite directions is *bad singular*. They then propose the following [7, Conjecture 1, `main.tex` lines 306–313], quoted character for character from the arXiv L^AT_EX source:

2020 *Mathematics Subject Classification*. 05C10, 05C30, 05C80.

Key words and phrases. random embedding, rotation system, signature, singular edge, facial walk, cubic graph, cycle double cover, counterexample.

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\begin{conjecture}\label{rnd-emb-conj}
In a random embedding of a bridgeless cubic graph  $G$ ,
\begin{enumerate}
\item The expected number of bad singular edges is  $\frac{m}{3}$ .
\item The expected number of good singular edges is  $\frac{m}{3}$ .
\item The expected number of regular edges is  $\frac{m}{3}$ .
\end{enumerate}
\end{conjecture}

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The sentence immediately preceding it, at `main.tex` lines 301–304, is the one that fixes m : “Let (π, λ) be a random embedding of a cubic graph G . Results of several computer experiments counting the number of singular edges in a random embedding of a cubic graph indicate that the expected number of bad singular edges is $\frac{m}{3}$, where m is the number of edges in G .”

That sentence fixes m , and it is all the source says about the probability space: no measure on embeddings is defined there or elsewhere in [7], so what follows depends on a choice of measure that we must make and state. We work with the uniform measure on signed rotation systems, which is the space the source’s own Mohar–Thomassen notation (π, λ) names, and we return in §5 to the uniform orientable model and to what is left open. Throughout, $n = |V(G)|$, $m = |E(G)| = 3n/2$, and

$$\Omega = \Omega(G) = \{(\pi, \lambda)\}, \quad |\Omega| = 2^n \cdot 2^m = 2^{n+m},$$

with the uniform measure: at each vertex of a cubic graph there are exactly 2 cyclic orders of the 3 darts, and λ ranges over $\{\pm 1\}^E$. Since n is even, $m/3 = n/2$ is an integer, so no crude integrality argument is available.

The quantifier over G is read as universal, i.e. one counterexample graph suffices. That is the source’s own use of the statement: its only consumer, Theorem 4.1 at `main.tex` lines 317–319, is proved at lines 320–322 by a first-moment argument applied to the single fixed G it is handed, which consumes $\mathbb{E}[\cdot]$ for that G and not an average over a family.

Two lines of work are adjacent to Conjecture 1 of [7] and settle no part of it. Bender and Richmond [1] bound the number of singular edges of an embedding of a 3-edge-connected graph in terms of the Euler characteristic of the surface — a deterministic worst-case bound, 0 on the sphere and 1 in the projective plane — with no random model and no expectation. The random-embedding papers of the source’s co-author and collaborators — Campion Loth and Mohar [2], and Campion Loth, Halasz, Mařařík, Mohar and řámal [3, 4] — compute expectations over uniformly random rotation systems, but of face counts only: none of them computes the expected number of bad singular, good singular or regular edges, and none addresses the $m/3$ split. The total-embedding-distribution literature, for instance Chen, Ou and Zou [5], enumerates signed rotation systems by genus rather than by edge class, and so does not record the class counts used below.

2. A DYADIC OBSTRUCTION

Theorem 1. *Let G be a finite connected edge-transitive cubic graph, and let Ω carry the uniform measure. Then for each of the three classes $i \in \{\text{bad singular, good singular, regular}\}$ one has $\mathbb{E}[\#\text{class } i] \neq m/3$; so each of the three parts of the conjecture fails for G in this model.*

Proof. Put $N = |\Omega| = 2^{n+m}$.

- (1) For each edge e and each class i , the number $A_{e,i}$ of $(\pi, \lambda) \in \Omega$ in which e lies in class i is a non-negative integer, so $\mathbb{P}[e \in \text{class } i] = A_{e,i}/N$ is dyadic.
- (2) Each $\sigma \in \text{Aut}(G)$ acts on Ω by $(\pi, \lambda) \mapsto (\pi^\sigma, \lambda^\sigma)$. Being a group action it is a bijection of the N systems, and it carries facial walks to facial walks preserving the number of traversals of an edge and their relative directions. Hence e lies in class i under (π, λ) if and only if $\sigma(e)$ lies in class i under the image, and $A_{e,i} = A_{\sigma(e),i}$.
- (3) Edge-transitivity makes all $A_{e,i}$ equal to a common A_i , so by linearity $\mathbb{E}[\#\text{class } i] = \sum_e A_{e,i}/N = m A_i/N$.
- (4) If $\mathbb{E}[\#\text{class } i] = m/3$ then $A_i/N = 1/3$, i.e. $3A_i = 2^{n+m}$, which is impossible because 3 divides no power of 2. This argument is run for each i separately. $\square$

Three remarks fix the strength of Theorem 1.

The divisibility must be applied per edge, never to the total: for a cubic graph $m = 3n/2$ is always divisible by 3, so $m/3 = n/2$ is an integer and “3 divides no power of 2” says nothing about $\mathbb{E}[\#\text{class } i] = m/3$ directly. Edge-transitivity is what converts the total into a per-edge statement, and without it the argument does not run.

“Bridgeless” is redundant in the hypothesis: a connected edge-transitive graph with a bridge has every edge a bridge, hence is a tree, and a tree has a leaf and so is not cubic.

The hypothesis is met by infinitely many graphs — infinitely many connected cubic arc-transitive graphs exist, the sextet construction of Biggs and Hoare [6] being one source — but the conclusion is quantitatively weak:

$$|\mathbb{E}[\#\text{class } i] - \frac{m}{3}| = \frac{m}{N} \left| A_i - \frac{N}{3} \right| \geq \frac{m}{3N},$$

since A_i is an integer and $3 \nmid N$; on K_4 this reads $1/512$. The floor improves to $1/32$ once one knows in addition that $2^n \mid A_i$, which holds because the switching action of Lemma 2 is free with all orbits of size 2^n and each class is constant on orbits: the nearest multiple of 16 to $N/3 = 1024/3$ is $16/3$ away, and $\frac{m}{N} \cdot \frac{16}{3} = 1/32$. By itself the theorem would license the reply “the exact equality is impossible, restate it asymptotically”; the computation of §3 shows the deviation on K_4 is ten times that floor.

3. THE K_4 COMPUTATION

Face tracing. Let G be cubic. Face tracing runs on the $4m$ arrival states (f, y, t) , where $f \in E(G)$, y is an endpoint of f and $t \in \{+1, -1\}$, under $\Phi = T \circ R$:

- R (rotate at y): from (f, y, t) , depart y along $g = \pi_y^t(f)$, keeping t ;
- T (traverse g from y): arrive at $(g, z, t \cdot \lambda(g))$, where z is the far end of g .

The reversal $\text{rev}(f, u, t) = (f, v, -t\lambda(f))$, with v the other end of f , is an involution satisfying $\text{rev} \circ \Phi \circ \text{rev} = \Phi^{-1}$, so facial walks correspond to pairs $\{C, \text{rev}(C)\}$ of Φ -cycles and the number of faces is $f = (\#\Phi\text{-orbits})/2$. Writing $A = (f, u, +1)$ and $B = (f, u, -1)$ for an edge $f = uv$, the definitions of [7] read

$$(1) \quad \begin{aligned} f \text{ good singular} &\iff A \sim_\Phi B, & f \text{ bad singular} &\iff A \sim_\Phi \text{rev}(B), \\ && \text{and } f \text{ regular otherwise,} \end{aligned}$$

and these are mutually exclusive and exhaustive (both would force $A \sim_\Phi \text{rev}(A)$, which is impossible). The conjecture's "1/3 each" shape is the guess that the three possible same-cycle relations among an edge's four tracing states are equally likely.

The switching lemma. Switching at a vertex v means replacing π_v by π_v^{-1} and flipping λ on the three edges at v ; this leaves the facial walks, hence all three edge classes, unchanged [7, `main.tex` lines 158–161]. Switching at $S \subseteq V$ therefore flips π on S and flips λ on the cut $\delta(S)$, and for fixed S the map $\lambda \mapsto \lambda \cdot \delta(S)$ is a bijection of $\{\pm 1\}^E$.

Lemma 2. $\mathbb{Z}_2^V$ acts on Ω by switching, the action is free with all orbits of size 2^n , each orbit meets $\{\pi = \pi_0\}$ in exactly one point, and all three edge classes are constant on orbits. Consequently, for every fixed rotation π_0 and every class i ,

$$\mathbb{E}_{(\pi, \lambda) \sim \Omega}[\# \text{class } i] = \mathbb{E}_{\lambda \sim \{\pm 1\}^E}[\# \text{class } i \mid \pi = \pi_0] \quad \text{exactly.}$$

Proof. Switching at S changes π at exactly the vertices of S , so the stabiliser of any point is trivial and orbits have size 2^n ; and the orbit of (π_0, λ) meets $\{\pi = \pi_0\}$ only at $S = \emptyset$. Class invariance is the vertex statement iterated over S . Averaging a class-constant function over Ω is therefore averaging it over the 2^m orbit representatives (π_0, λ) . $\square$

The count. Take K_4 on $\{0, 1, 2, 3\}$ with the six edges indexed $0, \dots, 5$ in the order

$$(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3),$$

so that dart $2i$ is the end of edge i at its smaller vertex and dart $2i + 1$ the other end. Let π_0 take, at each vertex, the three incident darts in increasing dart order; explicitly

$$\pi_0 : \quad 0 \mapsto (1, 2, 3), \quad 1 \mapsto (0, 2, 3), \quad 2 \mapsto (0, 1, 3), \quad 3 \mapsto (0, 1, 2)$$

as cyclic orders of neighbours. Tracing (1) at (π_0, λ) for each of the $2^6 = 64$ signatures λ classifies all six edges; the per-signature table — signature, number of faces, Euler genus $2 - (n - m + f)$, the six classes and the triple (bad, good, reg) — is recorded in the accompanying program `verify.py` rather than printed here, and any single row of it can be redone by hand. Over the 64 signatures the three class columns sum to 108, 138 and 138.

Theorem 3. *For $G = K_4$, which is edge-transitive, every edge is bad singular in exactly 288 of the 1024 signed rotation systems, good singular in exactly 368, and regular in exactly 368. Hence*

$$\mathbb{P}[e \text{ bad}] = \frac{288}{1024} = \frac{9}{32}, \quad \mathbb{P}[e \text{ good}] = \mathbb{P}[e \text{ regular}] = \frac{368}{1024} = \frac{23}{64},$$

and by linearity over the six edges

$$\mathbb{E}[\#\text{bad}] = \frac{27}{16} = 1.6875, \quad \mathbb{E}[\#\text{good}] = \mathbb{E}[\#\text{regular}] = \frac{69}{32} = 2.15625,$$

against the conjectured value $m/3 = 2$; in particular $2 \notin \{\frac{27}{16}, \frac{69}{32}, \frac{69}{32}\}$.

Proof. By Lemma 2 it suffices to average over the $2^6 = 64$ signatures at the single rotation π_0 ; each is classified by (1) from a trace of the 24 arrival states of K_4 , tabulated row by row in `verify.py`. The class column sums are 108, 138, 138, so by Lemma 2 the number of (system, edge) pairs over all of Ω with the edge bad singular is $\frac{1024}{64} \cdot 108 = 1728$, and $\frac{1024}{64} \cdot 138 = 2208$ for each of the other two classes. Now $\text{Aut}(K_4) = S_4$ is transitive on the six edges, so by step (2) of the proof of Theorem 1 the count $A_{e,i}$ is independent of e ; dividing by 6 gives 288, 368, 368 per edge. The probabilities and, by linearity, the expectations follow. $\square$

Corollary 4. *In the uniform model Ω , the three expectations of K_4 are all different from $m/3$, and no relabelling of the three classes changes that. Since K_4 is a simple, 3-connected, bridgeless cubic graph, each of the three parts of the conjecture fails on it under this measure.*

Proof. The conjecture asserts that all three class expectations equal $m/3$, so a relabelling permutes parts (1), (2), (3) and leaves the asserted set of equalities unchanged; by Theorem 3 the value $m/3 = 2$ is not an element of the multiset of the three expectations. $\square$

The orientable signatures. The system (π_0, λ) is orientable exactly when λ is a coboundary $\delta(S)$. The 16 vertex subsets of K_4 collapse 2-to-1 onto 8 distinct signatures, namely

$$\begin{array}{cccc} +++++ & -++++ & -++++ & +-+-- \\ +-+-- & +----- & -++++ & -++++ \end{array}$$

(in the same edge order as above). Their bad counts are 2, 3, 3, 3, 3, 2, 0, 2, with *good count 0 throughout*, exactly as [7, `main.tex` line 119] demands (“in orientable embeddings all singular edges are bad singular”); their Euler genera are all even, as orientability also demands. Neither fact was used in the tracing, so both test the good/bad convention against the source’s own assertion.

4. A SIGN-FLIP INVOLUTION

Line 304 of [7] says the expectation “seems to be the same for good singular edges and regular edges”. It is the same, exactly, for every edge of every loopless graph.

Theorem 5. *Let G be a graph without loops and $e \in E(G)$. Then flipping $\lambda(e)$ is a measure-preserving involution of Ω that exchanges the event $\{e \text{ regular}\}$ with $\{e \text{ good singular}\}$ and fixes $\{e \text{ bad singular}\}$; and it changes the number of faces by -1 , $+1$ and 0 respectively. Hence*

$$(2) \quad \mathbb{P}[e \text{ good}] = \mathbb{P}[e \text{ regular}] \quad \text{exactly,} \quad \mathbb{E}[\#\text{bad}] = m - 2\mathbb{E}[\#\text{good}],$$

so parts (2) and (3) of the conjecture are each equivalent to part (1), which is the aggregate assertion $\mathbb{E}[\#\text{bad}] = m/3$: equivalently, that a uniformly random edge of G is bad singular with probability $1/3$. No equality of the individual per-edge probabilities is asserted.

Proof. Write $e = uv$, $\lambda' = \lambda \cdot \mathbf{1}_e$, and let Φ, Φ' be the tracing permutations of (π, λ) and (π, λ') . Since $\lambda \mapsto \lambda'$ is a bijection of $\{\pm 1\}^E$ it is measure preserving, so only the classification has to be tracked.

Let S be the four states on e , and put $A = (e, u, +1)$, $B = (e, u, -1)$, $P = \text{rev}(A)$, $Q = \text{rev}(B)$, so that $\{P, Q\} = \{(e, v, +1), (e, v, -1)\}$ and $\text{rev}|_S = (AP)(BQ)$. Let $\tau = (AB)(PQ)$ be the involution of S negating the sign bit. Because e is not a loop, Φ leaves S at the first step, and a step of Φ arrives in S exactly when it traverses e ; that step is the only place $\lambda(e)$ is consulted. Hence Φ' agrees with Φ off those steps and satisfies $\Phi' = \tau \circ \Phi$ there, so: every Φ -orbit disjoint from S is also a Φ' -orbit, and the first-return maps $\sigma, \sigma' : S \rightarrow S$ of Φ, Φ' satisfy

$$(3) \quad \sigma' = \tau \circ \sigma.$$

Moreover $\#\Phi\text{-orbits} = \#\sigma\text{-cycles} + K$, where K counts the Φ -orbits disjoint from S , and the same with primes and the same K . Two states of S are Φ -equivalent iff they lie in one σ -cycle, so by (1), e is good singular iff A, B lie in one σ -cycle and bad singular iff A, Q do; and under λ' one has $\text{rev}'(A) = Q$, $\text{rev}'(B) = P$, so e is good singular iff A, B lie in one σ' -cycle and bad singular iff A, P do.

Now $\text{rev} \circ \Phi \circ \text{rev} = \Phi^{-1}$ and $\text{rev}(S) = S$ give $\text{rev} \sigma \text{rev} = \sigma^{-1}$ on S ; and no σ -cycle contains both A and $P = \text{rev}(A)$, nor both B and Q , since a Φ -cycle is never its own mirror. Enumerating the σ -cycle containing A under those two constraints leaves exactly three possibilities, and (3) determines each image:

$$\begin{array}{ll} \sigma = \text{id} & \sigma' = (AB)(PQ) \quad \text{regular} \rightarrow \text{good}, \\ \sigma = (AB)(PQ) & \sigma' = \text{id} \quad \text{good} \rightarrow \text{regular}, \\ \sigma = (AQ)(BP) & \sigma' = (AP)(BQ) \quad \text{bad} \rightarrow \text{bad}. \end{array}$$

(For the first case: if A is a fixed point of σ then $\sigma^{-1}(P) = \text{rev} \sigma \text{rev}(P) = \text{rev} \sigma(A) = P$ and likewise for Q , and B cannot be σ -joined to P without

violating $\sigma^{-1}(P) = P$; the other two cases are forced the same way from $\sigma(A) = B$ and $\sigma(A) = Q$.) This is the asserted exchange, and it also proves the trichotomy. Counting σ -cycles gives 4, 2, 2 respectively, so $f = (\#\sigma\text{-cycles} + K)/2$ changes by $-1, +1, 0$. The displayed identities follow from $\mathbb{P}[\text{bad}] + \mathbb{P}[\text{good}] + \mathbb{P}[\text{reg}] = 1$ and linearity over the m edges. $\square$

Neither Theorem 1 nor §3 uses (2); it is not needed to settle the conjecture, and is recorded only because it settles, as a theorem, a coincidence the source noticed empirically.

5. WHAT IS, AND IS NOT, SETTLED

The source paper keeps its theorem. The conjecture’s only consumer in [7] is Theorem 4.1 (`main.tex` lines 317–319): “Suppose that Conjecture 1 is true and let G be a bridgeless cubic graph. Then, there exists an embedding of G with at most $\frac{m}{3}$ singular edges.” The failure of the conjecture in the models treated here removes that hypothesis, so the proof at lines 320–322 no longer supports the conclusion. The conclusion itself is unaffected, and [7] already contains an unconditional proof of it in the referee remark at lines 324–325: take any perfect matching and extend the complementary 2-factor to a facial double cover; then only matching edges can be singular. A bridgeless cubic graph has a perfect matching [10], and a perfect matching of a cubic graph on n vertices has $n/2 = m/3$ edges, so the bound follows with no probabilistic input. What is lost is a conjecture and one redundant conditional proof, not a theorem.

The measure is not defined by the source. “Let (π, λ) be a random embedding” is the whole of it, so everything above depends on our choice of Ω and the conjecture is settled only under the models we name. Two uniform models were checked, and the three equalities fail under each: (i) uniform on the 2^{n+m} signed rotation systems, used above, and the space the source’s own Mohar–Thomassen notation names — sampling λ uniformly at a fixed π is not a further model but the same one in other coordinates, with the same induced distribution of edge classes and hence identical numbers, by Lemma 2; and (ii) uniform on the *orientable* embeddings, i.e. on the 2^n pure rotation systems, where on K_4 the eight coboundary signatures above give

$$\mathbb{E}[\#\text{bad}] = \frac{18}{8} = \frac{9}{4}, \quad \mathbb{E}[\#\text{good}] = 0, \quad \mathbb{E}[\#\text{regular}] = \frac{15}{4},$$

against $m/3 = 2$: parts (1) and (3) fail by $+\frac{1}{4}$ and $+\frac{7}{4}$, and part (2) fails degenerately by line 119. The sign of the deviation flips between (i) and (ii), so any statement of the form “ $\mathbb{E}[\#\text{bad}] \leq m/3$ ” must name its measure. A non-uniform or limiting measure is not covered: the source reports “several computer experiments”, it does not say what they sampled, and if they sampled some other space then what is settled here is the conjecture under each of the two uniform models above, not the statement about whatever was sampled there.

Only the exact constant is contradicted. The computed per-edge probabilities are all near $1/3$ — on K_4 , $\frac{9}{32} = 0.28125$ for bad and $\frac{23}{64} = 0.359375$ for good and for regular — and separating $\mathbb{E}[\#\text{bad}]$ from $m/3$ by sampling needs many samples. So what is contradicted is the exact constant $m/3$, not the near-uniform split the source observed. Whether $\mathbb{E}[\#\text{bad}]/m$ tends to $1/3$ along some family, and whether $\mathbb{E}[\#\text{bad}] < m/3$ always holds in the uniform signed-rotation model, are not addressed here; the latter is in any case false without the model qualifier, since the orientable model gives $9/4 > 2$ on K_4 .

Further limits. Theorem 1 covers only edge-transitive cubic graphs; for the rest only exhaustive counting is offered, on the seven graphs listed in §6. Nothing here proves the conjecture fails for *every* bridgeless cubic graph, though being universally quantified it needs only one counterexample. The infinitude of the connected cubic arc-transitive family is quoted from [6] and standard census material rather than verified here. The unproved Properties 6–9 of [7] are used nowhere above. The identification of the three computed expectations with the three parts of the conjecture is a reading of the source’s prose, made here and not verified by the program; nor does the program check that K_4 as it hard-codes it is simple and 3-connected, which is left to the reader. Finally, the program checks two conditions transcribed from the literature and used nowhere above: a bound of Bender and Richmond [1] on the number of singular edges of an embedding of a 3-edge-connected graph, in the form the authors of [7] themselves state it (their companion e-print arXiv:2511.07285, CDC.tex line 133), and the orientable genus distribution of K_4 given by Gross and Furst [8]. Neither the 1990 nor the 1987 paper was consulted here beyond its published abstract, so those two checks test the tracer against restatements only, and no claim about either is made.

6. VERIFICATION

The accompanying program `verify.py` (Python 3.9+, standard library only, exact integer and `Fraction` arithmetic, no floating-point decision) implements the face tracing of (1) from the definitions and re-derives every number above. It carries the 64-row table of K_4 signature data *as data* and compares it, row by row — signature, faces, Euler genus, six-letter class string and triple — with its own trace of the 24 arrival states of K_4 ; it then exhausts the full 2^{n+m} space of θ , K_4 , $K_{3,3}$ and the 3-prism, and all 2^m signatures at each of two fixed rotations of Q_3 , Wagner V_8 and Petersen, the latter licensed by Lemma 2; on each of those seven cells the three computed expectations all differ from $m/3$. That lemma is tested rather than assumed: the program switches at every vertex of every embedding of θ and K_4 , and at every vertex of a bounded sample on the other five cells, and it also checks directly that the per-edge counters are the same at every rotation it traces. Theorem 5 is likewise tested in its face-count form, on every (embedding, edge) pair of θ and K_4 . The program also checks the source’s line 119 on

every orientable embedding, checks that some embedding of K_4 does carry a good singular edge, so that the negative results are not vacuous, and reproduces row by row the two transcribed literature conditions recorded in §5, which are used nowhere in this paper. Here θ is the 3-dipole, a multigraph, and is a cell of this conjecture only if multigraphs count as bridgeless cubic, a reading the source neither grants nor denies; nothing above depends on it. The recorded run of the program is `verify.output.txt`.

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